

Literature Answer: Parasol Graphs Do Not Characterize Rolewicz Beta

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Status. `literature_already_answered`. The extracted problem from arXiv:1504.04250 has a negative answer in later literature.

Source question. In Baudier–Zhang, *(β)-distortion of some infinite graphs*, arXiv:1504.04250, Section “Metric characterization of asymptotic properties”, the authors prove that if a Banach space Y admits an equivalent norm with Rolewicz property (β) , then

$$\sup_{\ell \geq 1} c_Y(P_\ell^\omega) = \infty.$$

They then ask whether the converse holds:

$$Y \notin \langle(\beta)\rangle \implies \sup_{\ell \geq 1} c_Y(P_\ell^\omega) < \infty.$$

Later answer. Baudier–Causey–Dilworth–Kutzarova–Randrianarivony–Schlumprecht–Zhang, *On the geometry of the countably branching diamond graphs*, arXiv:1612.01984, Section 5 defines a family $\mathcal{G}^\omega$ of recursive finite-height bundle graphs and explicitly includes the countably branching parasol graphs P_k^ω . Their Theorem 5.1 states that if a Banach space Y admits an equivalent asymptotically midpoint uniformly convex norm, then

$$\sup_{k \in \mathbb{N}} c_Y(G_k^\omega) = \infty \quad ((G_k^\omega) \in \mathcal{G}^\omega).$$

Immediately after the proof, the authors point out that this theorem gives a negative answer to Problem 4.1 of Baudier–Zhang.

Concrete witness. The classical James space $\mathcal{J}$ is nonreflexive, so it cannot admit an equivalent norm with property (β) , because property (β) implies reflexivity. On the other hand, $\mathcal{J}$ has a 2-asymptotically uniformly convex norm; AUC implies AMUC. Applying Theorem 5.1 to the parasol sequence gives

$$\sup_{k \geq 1} c_{\mathcal{J}}(P_k^\omega) = \infty.$$

Thus $\mathcal{J} \notin \langle(\beta)\rangle$ while the parasol distortions are unbounded, so the converse asked in arXiv:1504.04250 is false.

Why this is not new. The supporting paper explicitly identifies the answer: it names the Baudier–Zhang problem and states that the theorem gives a negative answer. The James-space facts are included only to make the counterexample mechanism auditable.

References.

- F. P. Baudier and S. Zhang, (β) -distortion of some infinite graphs, arXiv:1504.04250.
- F. P. Baudier, R. Causey, S. Dilworth, D. Kutzarova, N. L. Randrianarivony, T. Schlumprecht, and S. Zhang, *On the geometry of the countably branching diamond graphs*, arXiv:1612.01984.
- F. P. Baudier and G. Lancien, *Geometric nonlinear functional analysis*, arXiv:2512.00817, Appendix on James spaces.